Daniel A. Delgado, PhD **Software Engineer | Spatial Computing, AI & Product Development**
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INDEPENDENT SOFTWARE PROJECTS

iCarus | Independent Software Engineer & UI Developer | 2026–Present
GitHub: github.com/xdaedalusx/icarus
• Built a native macOS media player from the ground up using SwiftUI, delivering audio/video playback, media downloads, customizable interfaces, and interactive controls.
• Engineered a modular skinning system that supports custom and image-based layouts, including Windows Media Player (.wmz) skin importing and conversion.
• Implemented a sandboxed JavaScript runtime for skin animations, events, and dynamic UI behavior, alongside a visual Skin Inspector and reusable components separating playback from presentation.

SPATIAL COMPUTING & AI PROJECTS | UNIVERSITY OF FLORIDA

Collaborative AR & Shared-Gaze Visualization | XR Software Engineer | 2024–2025
• Engineered interactive AR experiences combining eye tracking, real-time shared-gaze visualization, and spatial interface feedback to help collaborators understand each other's attention in shared environments.
• Iterated on visualization designs and interaction behavior using findings from three controlled studies involving 60+ participants, improving the alignment of interface behavior with collaboration needs.
• Translated user-performance and usability findings into implementation recommendations for collaborative AR interfaces.

ENKIx: Context-Aware AR Task Guidance | AR Developer & UI Designer | 2021–2024
• Built an interactive AR task-guidance application using Unity and C#/.NET, integrating ArUco-based computer vision to recognize objects and support context-aware interactions.
• Implemented contextual interfaces designed to guide users through complex tasks in constrained environments.

EMG-Based Prosthetic Simulation in Virtual Reality | Software Developer & Researcher | 2020–2023
• Built Unity-based VR simulations for prosthetic-device training, creating interactive environments for practicing EMG-based control.
• Implemented regression and deep neural network models for real-time EMG gesture classification, connecting machine-learning outputs to interactive VR experiences.
• Assessed usability and cognitive workload across virtual and physical training environments to inform software and interaction design.

EDUCATION **Ph.D., Computer Science** — University of Florida, Gainesville, FL | 2025

TECHNICAL SKILLS

Software Engineering: Swift, SwiftUI, C#/.NET, Python, JavaScript, HTML, CSS, Git
Spatial Computing: Unity, AR/VR development, HoloLens 2, eye tracking, computer vision, real-time visualization
AI & Machine Learning: Deep learning, neural networks, EMG gesture classification, TensorFlow, regression
Product & Interface Development: UI architecture, interaction design, prototyping, Figma, responsive layouts
Product Validation: Usability testing, user studies, experimental design, mixed-methods analysis
AI Development Tools: Claude, AI-assisted development, debugging, research and analysis